

Example 1: Scores

$\sigma_1 = 5$, $\sigma_2 = 6$, $n_1 = 40$, $n_2 = 35$, $\bar{x}_1 = 82$, $\bar{x}_2 = 78$, $\alpha =$

0 . 0 5 t w o - t a i l e d .

- $z = (82 - 78)/\sqrt{(5^2/40 + 6^2/35)} \approx 3.02$.
- Critical $z = \pm 1.960 \Rightarrow$ reject H_0 .
- 95% CI: $4 \pm 1.960 \cdot \sqrt{(5^2/40 + 6^2/35)} \Rightarrow (1.4, 6.6)$.

Example 2: Production Output (Right-Tailed)

$\sigma_1 = 3.2$, $\sigma_2 = 2.9$, $n_1 = 30$, $n_2 = 32$, $\bar{x}_1 = 105$, $\bar{x}_2 = 100$,

$\alpha =$ 0 . 0 1 r i g h t - t a i l e d .

- $z = (105 - 100)/\sqrt{(3.2^2/30 + 2.9^2/32)} \approx 6.03$.
- Critical $z(0.01) = 2.326 \Rightarrow$ reject H_0 .
- Effect size $d \approx (5)/\sqrt{((3.2^2 + 2.9^2)/2)} \approx 1.65$.